

Marmara University, 2020

Modeling and Discrete Simulation

Subject 6

Random-Number Generation

Mujdat Soyturk, Ph.D.
Associate Professor

Purpose & Overview



- Discuss the generation of random numbers.
- Introduce the subsequent testing for randomness:
  - Frequency test
  - Autocorrelation test.

Contents

- Random Numbers
- Techniques for Generating Random Numbers
- Tests for Random Numbers

Slides are derived from the course textbook "Banks, Carson, Nelson, Nicol. Discrete-Event System Simulation, 5th Edition, Prentice Hall, 2010" and from textbook's course slides.

RANDOM NUMBERS

The use of Random Numbers

- Random numbers are a necessary basic ingredient in the simulation of almost all discrete systems.
- Random numbers are used to generate event times and other random variables in the simulation languages.
- In the next lecture, you will learn how random numbers are used to generate a random variable with any desired probability distribution.
- In this lecture, you will learn how to generate random numbers.

Properties of Random Numbers

- A sequence of random numbers (sequentially generated by the same generator) **must have** two important statistical properties:
  - Uniformity
  - Independence.
- Random Number, R_i, must be independently drawn from a uniform distribution with *pdf*:

$$f(x) = \begin{cases} 1, & 0 \le x \le 1 \\ 0, & \text{otherwise} \end{cases}$$

$$E(R) = \int_0^1 x\,dx = \left.\frac{x^2}{2}\right|_0^1 = \frac{1}{2}$$

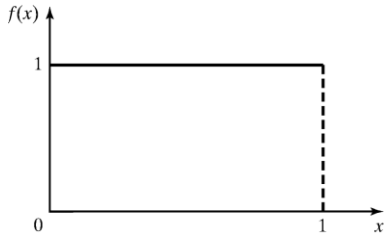


Figure: pdf for random numbers

Properties of Random Numbers

- The variance is:

$$V(R) = \int_0^1 x^2 dx - [E(R)]^2 = \frac{x^3}{3}\Big|_0^1 - \left(\frac{1}{2}\right)^2 = \frac{1}{3} - \frac{1}{4} = \frac{1}{12}$$

- Uniformity and independence property provide:
  - If the interval [0,1] is divided into *n* classes, or subintervals of equal length, the expected number of observations in each interval is *N/n*, where *N* is the total number of observations.
  - The probability of observing a value in a particular interval is independent of the previous values drawn.

Generation of Pseudo-Random Numbers

- "Pseudo", because generating numbers using a known method removes the potential for true randomness.
- Goal: To produce a sequence of numbers in [*0,1*] that simulates, or imitates, the ideal properties of uniform distribution and independence.
- Possible problems:
  - Generated RNs might not be uniformly distributed.
  - Generated RNs might be discrete-valued instead of continuous valued.
  - The mean of the generated RNs might be too high or too low.
  - The variance of the generated RNs might be too high or too low.
  - There might be dependence, e.g. as measured by autocorrelation.
- Any particular scheme have to be tested to detect departures from uniformity and independence.

Generation of Pseudo-Random Numbers

- Important considerations in random numbers (RN) routines:
  - Fast
  - Portable to different computers
  - Have sufficiently long cycle
  - Repeatable (completely independent of the system being simulated).
  - Closely approximate the ideal statistical properties of ***uniformity*** and ***independence***.

TECHNIQUES FOR GENERATING RANDOM NUMBERS

Techniques for Generating Random Numbers

- Linear Congruential Method (LCM).
- Combined Linear Congruential Generators (CLCG).
- Random-Number Streams.

Linear Congruential Method

- To produce a sequence of integers, X_1, X_2, … between *0* and *m-1* by following a recursive relationship:

$$X_{i+1} = (aX_i + c) \bmod m, \quad i = 0,1,2,...$$

The multiplier | The increment | The modulus

- **The selection of the values for *a*, *c*, *m*, and X_0 drastically affects** the **statistical properties and the cycle length**.
- The random integers are being generated [*0,m-1*], and to convert the integers to random numbers:

$$R_i = \frac{X_i}{m}, \quad i = 1,2,...$$

Example

- Use $X_0 = 27$, $a = 17$, $c = 43$, and $m = 100$.
- The X_i and R_i values are:

$X_1 = (17 * \mathbf{27} + 43) \bmod 100 = 502 \bmod 100 = 2,$ $R_1 = 0.02;$

$X_2 = (17 * \mathbf{2} + 43) \bmod 100 = 77,$ $R_2 = 0.77;$

$X_3 = (17 * \mathbf{77} + 43) \bmod 100 = 52,$ $R_3 = 0.52;$

…

Characteristics of a Good Generator

- **In addition to the uniformity and independence**, there are **secondary properties** that must be considered for the generated RNs.
- Maximum Density
  - Such that the values assumed by R_i, $i = 1,2,\ldots$, leave no large gaps on *[0,1]*
  - Problem: Instead of continuous, each R_i is discrete
  - Solution: a very large integer for modulus m
    - Approximation appears to be of little consequence
- Maximum Period
  - To achieve maximum density and avoid cycling.
  - Achieve by: proper choice of a, c, m, and X_0.
- Most digital computers use a binary representation of numbers
  - Speed and efficiency are aided by a modulus, m, to be (or close to) a power of *2*.

Characteristics of a Good Generator

i	X_i	X_i	X_i	X_i
0	1	2	3	4
1	13	26	39	52
2	41	18	59	36
3	21	42	63	20
4	17	34	51	4
5	29	58	23	
6	57	50	43	
7	37	10	47	
8	33	2	35	
9	45		7	
10	9		27	
11	53		31	
12	49		19	
13	61		55	
14	25		11	
15	5		15	
16	1		3	

cycle

multiplicative generator with a=13, m=2^6=64, (c=0 in multiplicative generators)

Combined Linear Congruential Generators

- Reason: Longer period generator is needed because of the increasing complexity of simulated systems.
- Approach: Combine two or more multiplicative congruential generators.

- Let $X_{i,1}, X_{i,2}, \ldots, X_{i,k}$ be the ith output from k different multiplicative congruential generators.
  - The jth generator:
    - Has prime modulus m_j and multiplier a_j, so the period is $m_j - 1$
    - Produces integers $X_{i,j}$ is approx ~ Uniform on integers in $[1, m_j - 1]$
    - $W_{i,j} = X_{i,j} - 1$ is approx ~ Uniform on integers in $[0, m-2]$

Combined Linear Congruential Generators

– Suggested form:

$$X_i = \left(\sum_{j=1}^{k} (-1)^{j-1} X_{i,j} \right) \bmod (m_1 - 1)$$

The coefficient: Performs the subtraction $X_{i,1-1}$

$$\text{Hence, } R_i = \begin{cases} \dfrac{X_i}{m_1}, & X_i > 0 \\ \dfrac{m_1 - 1}{m_1}, & X_i = 0 \end{cases}$$

- The maximum possible period is:

$$P = \frac{(m_1 - 1)(m_2 - 1)\ldots(m_k - 1)}{2^{k-1}}$$

Combined Linear Congruential Generators

- Example: For 32-bit computers, L'Ecuyer [1988] suggests combining $k = 2$ generators with $m_1 = 2{,}147{,}483{,}563$, $a_1 = 40{,}014$, $m_2 = 2{,}147{,}483{,}399$ and $a_2 = 20{,}692$. The algorithm becomes:

  Step 1: Select seeds
  - $X_{1,0}$ in the range $[1, 2{,}147{,}483{,}562]$ for the 1st generator
  - $X_{2,0}$ in the range $[1, 2{,}147{,}483{,}398]$ for the 2nd generator.

  Step 2: For each individual generator,
  $X_{1,j+1} = 40{,}014\ X_{1,j} \bmod 2{,}147{,}483{,}563$
  $X_{2,j+1} = 40{,}692\ X_{1,j} \bmod 2{,}147{,}483{,}399$.

  Step 3: $X_{j+1} = (X_{1,j+1} - X_{2,j+1}) \bmod 2{,}147{,}483{,}562$.

  Step 4: Return

$$R_{j+1} = \begin{cases} \dfrac{X_{j+1}}{2{,}147{,}483{,}563}, & X_{j+1} > 0 \\ \dfrac{2{,}147{,}483{,}562}{2{,}147{,}483{,}563}, & X_{j+1} = 0 \end{cases}$$

  Step 5: Set $j = j+1$, go back to step 2.

– Combined generator has period: $(m_1 - 1)(m_2 - 1)/2 \approx 2 \times 10^{18}$

Random-Numbers Streams

- The seed for a linear congruential random-number generator:
  - Is the integer value X_0 that initializes the random-number sequence.
  - Any value in the sequence can be used to "seed" the generator.
- A random-number stream:
  - Refers to a starting seed taken from the sequence $X_0, X_1, \ldots, X_P$.
  - If the streams are *b* values apart, then stream *i* could defined by starting seed:
    $$S_i = X_{b(i-1)} \quad \text{for } i = 1,2,\ldots,\lfloor P/b \rfloor$$
  - Older generators: $b = 10^5$; Newer generators: $b = 10^{37}$.
- A single random-number generator with ***k*** streams can act like ***k*** distinct virtual random-number generators
- To compare two or more alternative systems.
  - Advantageous to dedicate portions of the pseudo-random number sequence to the same purpose in each of the simulated systems.
  - This is considered as synchronization.
  - Provides fairer comparison.

TESTS FOR RANDOM NUMBERS

Tests for Random Numbers



- Two categories:
  - Testing for uniformity:

    H_0: $R_i \sim U[0,1]$

    H_1: $R_i \not\sim U[0,1]$

    - Failure to reject the null hypothesis, H_0, means that evidence of non-uniformity has not been detected.
    - ***Frequency tests*** are used for this purpose.
  - Testing for independence:

    H_0: $R_i \sim$ independently

    H_1: $R_i \not\sim$ independently

    - Failure to reject the null hypothesis, H_0, means that evidence of dependence has not been detected.
    - ***Autocorrelation tests*** are used for this purpose.
- **Level of significance α**, the probability of rejecting H_0 when it is true:

  $\alpha = P(\text{reject } H_0 | H_0 \text{ is true})$

Tests for Random Numbers



- When to use these tests:
  - If a well-known simulation languages or random-number generators is used, it is probably unnecessary to test
  - If the generator is not explicitly known or documented, e.g., spreadsheet programs, symbolic/numerical calculators, tests should be applied to many sample numbers.
- Types of tests:
  - Theoretical tests: evaluate the choices of ***m***, ***a***, and ***c*** without actually generating any numbers
  - Empirical tests: applied to actual sequences of numbers produced. Our emphasis.

Frequency Tests

- Test of uniformity
- Two different methods:
  - Kolmogorov-Smirnov test
  - Chi-square test
- Both tests the degree of agreement between the distribution of a sample of generated RNs and the theoretical uniform distribution.

Kolmogorov-Smirnov Test

- Compares the continuous *cdf*, *F(x)*, of the uniform distribution with the empirical *cdf*, $S_N(x)$, of the *N* sample observations.
  - We know: $F(x) = x, \quad 0 \le x \le 1$
  - If the sample from the RN generator is $R_1, R_2, \ldots, R_N$, then the empirical *cdf*, $S_N(x)$ is:

$$S_N(x) = \frac{\text{number of } R_1, R_2, \ldots, R_n \text{ which are} \le x}{N}$$

- Based on the statistic: $D = \max |F(x) - S_N(x)|$ *(D: deviation)*
  - Sampling distribution of *D* is known (a function of *N*, tabulated in Table A.8.)
- A more powerful test, recommended.

Kolmogorov-Smirnov Test

Example: Suppose *5* generated numbers are *0.44, 0.81, 0.14, 0.05, 0.93*.

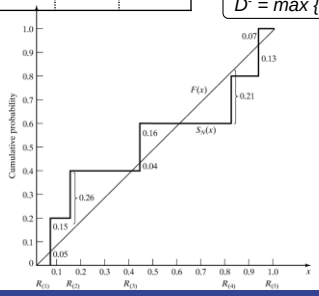
Step 1:	$R_{(i)}$	0.05	0.14	0.44	0.81	0.93	Arrange $R_{(i)}$ from smallest to largest
	i/N	0.20	0.40	0.60	0.80	1.00	
Step 2:	$i/N - R_{(i)}$	0.15	0.26	0.16	-	0.07	$D^+ = max\{i/N - R_{(i)}\}$
	$R_{(i)} - (i-1)/N$	0.05	-	0.04	0.21	0.13	$D^- = max\{R_{(i)} - (i-1)/N\}$

Step 3: $D = max(D^+, D^-) = 0.26$

Step 4: For $\alpha = 0.05$,

$D_\alpha = 0.565 > D$

Hence, H_0 is not rejected.



Chi-square test

- Chi-square test uses the sample statistic:

$$\chi_0^2 = \sum_{i=1}^{n} \frac{(O_i - E_i)^2}{E_i}$$

n is the # of classes

E_i is the expected # in the ith class

O_i is the observed # in the ith class

  - Approximately the chi-square distribution with ***n-1*** degrees of freedom (where the critical values are tabulated in Table A.6)
  - For the uniform distribution, E_i, the expected number in the each class is:

$$E_i = \frac{N}{n}, \quad \text{where N is the total \# of observation}$$

- *n* and *N* should be chosen so that each $E_i \geq 5$
- Valid only for large samples, e.g. N ≥ 50

Chi-square test

Example 7.7

Use the chi-square test with $\alpha = 0.05$ to test for whether the data shown next are uniformly distributed. Table 7.3 contains the essential computations. The test uses $n = 10$ intervals of equal length, namely [0, 0.1), [0.1, 0.2), ..., [0.9, 1.0). The value of χ_0^2 is 3.4. This is compared with the critical value $\chi^2_{0.05,9} = 16.9$ from Table A.6. Since χ_0^2 is much smaller than the tabulated value of $\chi^2_{0.05,9}$, the null hypothesis of a uniform distribution is not rejected.

0.34	0.90	0.25	0.89	0.87	0.44	0.12	0.21	0.46	0.67
0.83	0.76	0.79	0.64	0.70	0.81	0.94	0.74	0.22	0.74
0.96	0.99	0.77	0.67	0.56	0.41	0.52	0.73	0.99	0.02
0.47	0.30	0.17	0.82	0.56	0.05	0.45	0.31	0.78	0.05
0.79	0.71	0.23	0.19	0.82	0.93	0.65	0.37	0.39	0.42
0.99	0.17	0.99	0.46	0.05	0.66	0.10	0.42	0.18	0.49
0.37	0.51	0.54	0.01	0.81	0.28	0.69	0.34	0.75	0.49
0.72	0.43	0.56	0.97	0.30	0.94	0.96	0.58	0.73	0.05
0.06	0.39	0.84	0.24	0.40	0.64	0.40	0.19	0.79	0.62
0.18	0.26	0.97	0.88	0.64	0.47	0.60	0.11	0.29	0.78

Chi-square test

Example 7.7

Use the chi-square test with $\alpha = 0.05$ to test for whether the data shown next are uniformly distributed. Table 7.3 contains the essential computations. The test uses $n = 10$ intervals of equal length, namely [0, 0.1), [0.1, 0.2), ..., [0.9, 1.0). The value of χ_0^2 is 3.4. This is compared with the critical value $\chi^2_{0.05,9} = 16.9$ from Table A.6. Since χ_0^2 is much smaller than the tabulated value of $\chi^2_{0.05,9}$, the null hypothesis of a uniform distribution is not rejected.

0.34	0.90	0.25	0.89	0.87	0.44	0.12	0.21	0.46	0.67
0.83	0.76	0.79	0.64	0.70	0.81	0.94	0.74	0.22	0.74
0.96	0.99	0.77	0.67	0.56	0.41	0.52	0.73	0.99	0.02
0.47	0.30	0.17	0.82	0.56	0.05	0.45	0.31	0.78	0.05
0.79	0.71	0.23	0.19	0.82	0.93	0.65	0.37	0.39	0.42
0.99	0.17	0.99	0.46	0.05	0.66	0.10	0.42	0.18	0.49
0.37	0.51	0.54	0.01	0.81	0.28	0.69	0.34	0.75	0.49
0.72	0.43	0.56	0.97	0.30	0.94	0.96	0.58	0.73	0.05
0.06	0.39	0.84	0.24	0.40	0.64	0.40	0.19	0.79	0.62
0.18	0.26	0.97	0.88	0.64	0.47	0.60	0.11	0.29	0.78

Chi-square test

Example 7.7

Use the chi-square test with $\alpha = 0.05$ to test for whether the data shown next are uniformly distributed. Table 7.3 contains the essential computations. The test uses $n = 10$ intervals of equal length, namely [0, 0.1), [0.1, 0.2), ..., [0.9, 1.0). The value of χ_0^2 is 3.4. This is compared with the critical value $\chi^2_{0.05,9} = 16.9$ from Table A.6. Since χ_0^2 is much smaller than the tabulated value of $\chi^2_{0.05,9}$, the null hypothesis of a uniform distribution is not rejected.

0.34	0.90	0.25	0.89	0.87	0.44	0.12	0.21	0.46	0.67
0.83	0.76	0.79	0.64	0.70	0.81	0.94	0.74	0.22	0.74
0.96	0.99	0.77	0.67	0.56	0.41	0.52	0.73	0.99	0.02
0.47	0.30	0.17	0.82	0.56	0.05	0.45	0.31	0.78	0.05
0.79	0.71	0.23	0.19	0.82	0.93	0.65	0.37	0.39	0.42
0.99	0.17	0.99	0.46	0.05	0.66	0.10	0.42	0.18	0.49
0.37	0.51	0.54	0.01	0.81	0.28	0.69	0.34	0.75	0.49
0.72	0.43	0.56	0.97	0.30	0.94	0.96	0.58	0.73	0.05
0.06	0.39	0.84	0.24	0.40	0.64	0.40	0.19	0.79	0.62
0.18	0.26	0.97	0.88	0.64	0.47	0.60	0.11	0.29	0.78

Chi-square test

Example 7.7

Use the chi-square test with $\alpha = 0.05$ to test for whether the data shown next are uniformly distributed. Table 7.3 contains the essential computations. The test uses $n = 10$ intervals of equal length, namely [0, 0.1), [0.1, 0.2), ..., [0.9, 1.0). The value of χ_0^2 is 3.4. This is compared with the critical value $\chi^2_{0.05,9} = 16.9$ from Table A.6. Since χ_0^2 is much smaller than the tabulated value of $\chi^2_{0.05,9}$, the null hypothesis of a uniform distribution is not rejected.

0.34	0.90	0.25	0.89	0.87	0.44	0.12	0.21	0.46	0.67
0.83	0.76	0.79	0.64	0.70	0.81	0.94	0.74	0.22	0.74
0.96	0.99	0.77	0.67	0.56	0.41	0.52	0.73	0.99	0.02
0.47	0.30	0.17	0.82	0.56	0.05	0.45	0.31	0.78	0.05
0.79	0.71	0.23	0.19	0.82	0.93	0.65	0.37	0.39	0.42
0.99	0.17	0.99	0.46	0.05	0.66	0.10	0.42	0.18	0.49
0.37	0.51	0.54	0.01	0.81	0.28	0.69	0.34	0.75	0.49
0.72	0.43	0.56	0.97	0.30	0.94	0.96	0.58	0.73	0.05
0.06	0.39	0.84	0.24	0.40	0.64	0.40	0.19	0.79	0.62
0.18	0.26	0.97	0.88	0.64	0.47	0.60	0.11	0.29	0.78

Chi-square test

Example 7.7

Use the chi-square test with $\alpha = 0.05$ to test for whether the data shown next are uniformly distributed. Table 7.3 contains the essential computations. The test uses $n = 10$ intervals of equal length, namely [0, 0.1), [0.1, 0.2), ..., [0.9, 1.0). The value of χ_0^2 is 3.4. This is compared with the critical value $\chi^2_{0.05,9} = 16.9$ from Table A.6. Since χ_0^2 is much smaller than the tabulated value of $\chi^2_{0.05,9}$, the null hypothesis of a uniform distribution is not rejected.

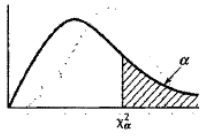
0.34	0.90	0.25	0.89	0.87	0.44	0.12	0.21	0.46	0.67
0.83	0.76	0.79	0.64	0.70	0.81	0.94	0.74	0.22	0.74
0.96	0.99	0.77	0.67	0.56	0.41	0.52	0.73	0.99	0.02
0.47	0.30	0.17	0.82	0.56	0.05	0.45	0.31	0.78	0.05
0.79	0.71	0.23	0.19	0.82	0.93	0.65	0.37	0.39	0.42
0.99	0.17	0.99	0.46	0.05	0.66	0.10	0.42	0.18	0.49
0.37	0.51	0.54	0.01	0.81	0.28	0.69	0.34	0.75	0.49
0.72	0.43	0.56	0.97	0.30	0.94	0.96	0.58	0.73	0.05
0.06	0.39	0.84	0.24	0.40	0.64	0.40	0.19	0.79	0.62
0.18	0.26	0.97	0.88	0.64	0.47	0.60	0.11	0.29	0.78

Chi-square test

Interval	O_i	E_i	$O_i - E_i$	$(O_i - E_i)^2$	$\frac{(O_i - E_i)^2}{E_i}$
1	8	10	−2	4	0.4
2	8	10	−2	4	0.4
3	10	10	0	0	0.0
4	9	10	−1	1	0.1
5	12	10	2	4	0.4
6	8	10	−2	4	0.4
7	10	10	0	0	0.0
8	14	10	4	16	1.6
9	10	10	0	0	0.0
10	11	10	1	1	0.1
	100	100	0		3.4

Chi-square test

Table A.6 Percentage Points of The Chi-Square Distribution with ν Degrees of Freedom



α=0.05

ν	$\chi^2_{0.005}$	$\chi^2_{0.01}$	$\chi^2_{0.025}$	$\chi^2_{0.05}$	$\chi^2_{0.10}$
1	7.88	6.63	5.02	3.84	2.71
2	10.60	9.21	7.38	5.99	4.61
3	12.84	11.34	9.35	7.81	6.25
4	14.96	13.28	11.14	9.49	7.78
5	16.7	15.1	12.8	11.1	9.2
6	18.5	16.8	14.4	12.6	10.6
7	20.3	18.5	16.0	14.1	12.0
8	22.0	20.1	17.5	15.5	13.4
df=9 → 9	23.6	21.7	19.0	16.9	14.7
10	25.2	23.2	20.5	18.3	16.0
11	26.8	24.7	21.9	19.7	17.3
12	28.3	26.2	23.3	21.0	18.5

Tests for Autocorrelation

0.12	0.01	0.23	0.28	0.89	0.31	0.64	0.28	0.83	0.93
0.99	0.15	0.33	0.35	0.91	0.41	0.60	0.27	0.75	0.88
0.68	0.49	0.05	0.43	0.95	0.58	0.19	0.36	0.69	0.87

Tests for Autocorrelation

0.12	0.01	0.23	0.28	0.89	0.31	0.64	0.28	0.83	0.93
0.99	0.15	0.33	0.35	0.91	0.41	0.60	0.27	0.75	0.88
0.68	0.49	0.05	0.43	0.95	0.58	0.19	0.36	0.69	0.87

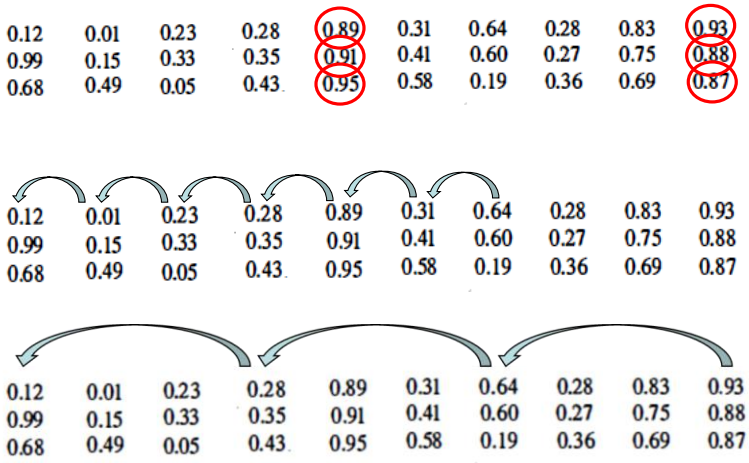
Tests for Autocorrelation

0.12	0.01	0.23	0.28	0.89	0.31	0.64	0.28	0.83	0.93
0.99	0.15	0.33	0.35	0.91	0.41	0.60	0.27	0.75	0.88
0.68	0.49	0.05	0.43	0.95	0.58	0.19	0.36	0.69	0.87



0.12	0.01	0.23	0.28	0.89	0.31	0.64	0.28	0.83	0.93
0.99	0.15	0.33	0.35	0.91	0.41	0.60	0.27	0.75	0.88
0.68	0.49	0.05	0.43	0.95	0.58	0.19	0.36	0.69	0.87

Tests for Autocorrelation



Tests for Autocorrelation

- Testing the autocorrelation between every ℓ numbers (ℓ is a.k.a. the lag), starting with the ith number
  - The autocorrelation $\rho_{i\ell}$ between numbers: R_i, $R_{i+\ell}$, $R_{i+2\ell}$, $R_{i+(M+1)\ell}$
  - M is the largest integer such that $i+(M+1)\ell \leq N$
- Hypothesis:

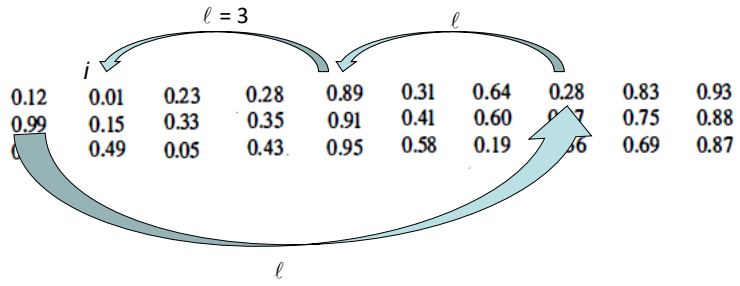
$$H_0: \rho_{i\ell} = 0, \quad \text{if numbers are independent}$$
$$H_1: \rho_{i\ell} \neq 0, \quad \text{if numbers are dependent}$$

- If the values are uncorrelated:
  - For large values of M, the distribution of the estimator of $\rho_{i\ell}$, denoted $\hat{\rho}_{i\ell}$ is approximately normal.

Tests for Autocorrelation

- 30 numbers is a rather small sample size on which to reject a random number generator, but the notion is that numbers in the sequence might be related.
- A method for discovering whether such a relationship exists
  - Autocorrelation test
  - Between high numbers, low numbers, or high to low or vice versa
  - Between every ℓ numbers

Tests for Autocorrelation



ℓ : (lag)
i : position of the starting value

autocorrelation $\rho_{i\ell}$
autocorrelation $\rho_{2,3}$

Tests for Autocorrelation

- Test statistics is:

$$Z_0 = \frac{\hat{\rho}_{i\ell}}{\sigma_{\hat{\rho}_{i\ell}}}$$

  - Z_0 is distributed normally with mean = *0* and variance = *1*, and:

$$\hat{\rho}_{i\ell} = \frac{1}{M+1}\left[\sum_{k=0}^{M} R_{i+k\ell} R_{i+(k+1)\ell}\right] - 0.25$$

$$\sigma_{\hat{\rho}_{i\ell}} = \frac{\sqrt{13M+7}}{12(M+1)}$$

- If $\rho_{i\ell} > 0$, the subsequence has positive autocorrelation
  - High random numbers tend to be followed by high ones, and vice versa.
- If $\rho_{i\ell} < 0$, the subsequence has negative autocorrelation
  - Low random numbers tend to be followed by high ones, and vice versa.

Example

- Test whether the *3rd*, *8th*, *13th*, and so on, for the following output.

0.12	0.01	0.23	0.28	0.89	0.31	0.64	0.28	0.83	0.93
0.99	0.15	0.33	0.35	0.91	0.41	0.60	0.27	0.75	0.88
0.68	0.49	0.05	0.43	0.95	0.58	0.19	0.36	0.69	0.87

  - Hence, $\alpha = 0.05$, $i = 3$, $\ell = 5$, $N = 30$, and M = 4

$$\rho_{35} = \frac{1}{4+1}\left[\begin{array}{l}(0.23)(0.28)+(0.28)(0.33)+(0.33)(0.27)\\+(0.27)(0.05)+(0.05)(0.36)\end{array}\right] - 0.25$$

$$= -0.1945$$

$$\hat{\sigma}_{\rho_{35}} = \frac{\sqrt{13(4)+7}}{12(4+1)} = 0.128$$

$$Z_0 = -\frac{0.1945}{0.1280} = -1.516$$

  - From Table A.3, $z_{0.025} = 1.96$. Hence, the hypothesis is not rejected.

Shortcomings



- The test is not very sensitive for small values of *M,* particularly when the numbers being tests are on the low side.
- Problem when "fishing" for autocorrelation by performing numerous tests:
  - If $\alpha = 0.05$, there is a probability of 0.05 of rejecting a true hypothesis.
  - If 10 independence sequences are examined,
    - The probability of finding no significant autocorrelation, by chance alone, is $0.95^{10} = 0.60$.
    - Hence, the probability of detecting significant autocorrelation when it does not exist = *40%*

Summary



- In this lecture, we described:
  - Generation of random numbers
  - Testing for uniformity and independence
- Caution:
  - Even with generators that have been used for years, some of which still in used, are found to be inadequate.
  - This lecture provides only the basic
  - Also, even if generated numbers pass all the tests, some underlying pattern might have gone undetected.